Project Overview

Business Problem:

A telecom company is experiencing customer attrition, leading to revenue loss and business risk.

The objective is to identify:

- Churn drivers
- Revenue at risk
- High-risk customer segments
- Retention opportunities

Dataset Overview:

Dataset: Telco Customer Churn Dataset

Records: 7043 Customers

Fields: 21 Columns

Key Business Questions:

1. What is the current churn rate?
2. How many active customers remain?
3. How much revenue is at risk?
4. Which customer segments are most likely to churn?
5. What customer characteristics drive churn?
6. Which retention opportunities should leadership prioritize?

Data Modeling:

Facts

- Monthly Charges
- Total Charges
- Tenure

Dimensions

- Contract Type
- Internet Service
- Phone Service
- Payment Method
- Streaming Services
- Senior Citizen
- Dependents
- Customer Status

Measures Created

Active Customers - Number of non-churned customers

Churn Rate - Percentage of customers who left

Revenue at Risk - Potential revenue associated with churned customers

Revenue at Risk % - Revenue at risk compared with total customer revenue

Key Insights and Findings:

1. Contract type is the strongest churn driver. Customers on month-to-month contracts exhibit a 74.55% churn rate compared to only 2.91% for two-year contracts.

2. Early tenure customers represent the highest risk segment. Customers within their first year show a 90.25 % churn rate, indicating onboarding and retention challenges during the initial customer lifecycle.

3. Electronic check customers are significantly more likely to churn. This segment shows the highest churn ra te (82.77%) and should be targeted for retention analysis.

4. Fiber optic customers contribute the highest proportion of churned users. Additional investigation is reco mmended to determine whether service quality, pricing, or customer expectations are contributing factors.

5. Current churn levels are creating approximately $139K in monthly revenue loss, representing 30.5% reven ue at risk.